A Good Fit to the Missing Mass Problem in Galaxies and Clusters of Galaxies

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Abstract. Precise measurements of the distribution of matter in galaxies and clusters of galaxies is essential to resolving the missing mass problem. A universal fitting formula for the distribution of dark matter is revealed to fit galaxy rotation curves of a sample of 29 galaxies, including high and low surface brightness and dwarf galaxies in the Ursa Major filament, without introducing the cusp problem of singular halo profiles. The asymptotic density profile is well described by a steep power-law, $\rho(r) \propto r^{-3}$, at large distances from the center of the galaxy, and an increasingly shallow power-law toward the center of the galaxy approaching a rarified, constant density core, where visible baryons dominate the Newtonian force law. The same universal fitting formula is revealed to fit cluster mass profiles of a sample of 10 X-ray clusters of galaxies. The variation of the ratio of the inferred Newtonian dynamic mass to the observed baryon mass is compared across the scales, demonstrating the crucial role played by baryons, and defining what a good theory must achieve in order to provide a good fit to the missing mass problem, and provide a test bed for the predictions of general relativity, constraining modified theories of gravity.

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1. INTRODUCTION

Enormous strides have been taken to understand the growth of the large scale structure of the Universe from studying the anisotropy power spectrum of the cosmic microwave background [1, 2], and cross-correlating these measurements with the baryon acoustic oscillations imprinted on the spatial distribution of the clustering of galaxies [3, 4, 5], and with the luminosity distances of Type Ia Supernovae [6]. Flat (or nearly-flat) cosmological models with a mixture of radiation, ordinary baryonic matter, cold collisionless dark matter and cosmological constant (or quintessence) and a nearly scale-invariant adiabatic spectrum of primordial density fluctuations provide remarkable fits to large scale ($\gg$ 1 Mpc) observations. While substantial progress has been made in constraining the flat-ΛCDM cosmological parameters to within a few percentages [7], there is still a lack of understanding of the evolution of the baryonic component, which is deeply connected with the formation and evolution of structure at various scales [8]. There remains a large amount of data on galactic and sub-galactic scales ($\ll$ 100 kpc) which require universal explanation [9]. As computing power increases, and hydrodynamical simulations improve, it is unclear whether the inner dark matter density converges to a single power-law form, rather than becoming progressively shallower in slope at smaller radii [10, 11]. Analysis of galaxy rotation curves for late-type spiral galaxies indicate that they must have shallow, non-cusped dark matter profiles, most noticeable for dwarf galaxies [12, 13, 14]. Analysis of X-ray clusters of galaxies indicate inner slopes for the

total mass profiles ranging from steep to shallow, most shallow for the smallest concentration of the gas mass fraction, [14, 15].

Our current knowledge of the baryon component comes from multiwavelength observations, including the 21 cm emission for neutral HI, infrared photometry of the interstellar medium, X-ray spectral imaging for the $> 10^7 K$ intracluster medium, radio synchrotron emission for high-energy plasma, and the velocity and turbulence fields of galaxies and clusters of galaxies. Combined multiwavelength observations culminate in high-resolution galaxy rotation curves and X-ray cluster images [14]. Outstanding scientific questions include:

- ► How does the Universe evolve from large linear scales dominated by dark matter to the non-linear scales of galaxies and clusters, where baryons and dark matter both play important, interacting, roles?
- ► Do relaxed structures exhibit steep or increasingly shallow density profiles in the central regions due to cores of dense baryons?
- ► What is the dark matter self-interaction coupling strength? Can the nature of dark matter be sufficiently constrained to predict a candidate particle from the list of standard model extensions?
- ► In the absence of dominant dark matter, is there a modification to the theory of gravity which can solve the missing mass problem, at all scales, universally?

To address these questions, the technique of imaging dark matter halos is presented in §2, and we present what is known about the global mass to light ratio in §3 by considering a sample of 19 high surface brightness (HSB) spiral galaxies and 10 low surface brightness (LSB) spiral galaxies from the Ursa Major filament in §3.1 and a sample of 10 X-ray clusters of galaxies in §3.2. Conclusions are drawn in §4.

2. IMAGING DARK MATTER HALOS

Ostriker et al. [16] argued that the masses of ordinary galaxies – found by assuming a constant mass-to-light ratio – may have been underestimated by a factor of 10; but that the galaxy rotation curves in the inner regions provide almost no information about the exterior halo masses. Upon application of a Newtonian force law,

$$a(r) = -\frac{G_N M(r)}{r^2}, \tag{1}$$

where G_N is Newton's constant measured experimentally[1], one may obtain the Newtonian dynamic mass, $M(r)$, which is the mass interior to the sphere of radius, r, needed to support the galaxy rotation curve.

Ostriker et al. [16] observed that although the surface luminosity profiles, $L(r)$, do appear to be convergent, the Newtonian dynamic masses, $M(r)$, diverge with r either

[1] $G_N = 6.67428(67) \times 10^{-11} \mathrm{m}^3 \mathrm{kg}^{-1} \mathrm{s}^{-2}$ [NIST 2006 CODATA value]

weakly (logarithmic) or strongly (linear) depending on the method of measurement, and concluded that within local giant spiral galaxies,

$$M(r) \propto r \qquad \text{for 20 kpc } \leq r \leq 500 \text{ kpc}. \tag{2}$$

High-resolution galaxy rotation curves, for both HSB and LSB spiral galaxies, can be fitted to the total mass density profile

$$\rho \propto r^{-\gamma(r)}. \tag{3}$$

The fits grossly reproduce the observed flat rotation curves with $\langle\gamma\rangle = 2$, corresponding to mass profiles which diverge linearly with distance. The divergent mass-to-light ratio necessitates the existence of giant halos surrounding ordinary galaxies of dark matter – the implied density distribution similar to isothermal gas spheres in the outer parts, [17]

$$\rho(r) = \frac{\rho_0 r_c^2}{r^2 + r_c^2}, \tag{4}$$

where r_c is the core radius and ρ_0 is the central dark matter density. In the limit of small $r \ll r_c$, the isothermal sphere model approaches a constant density core. Spherically integrating the constant density core model of Equation (4) one obtains a simple fitting formula for the mass of dark matter,

$$M(r) = 4\pi\rho_0 r_c^3 \left\{ \frac{r}{r_c} - \tan^{-1}(r/r_c) \right\}, \tag{5}$$

which diverges with the behavior of Equation (2), for $r \gg r_c$.

In search of a universal description of collisionless dark matter, Navarro et al. [18, 19] provided power-law fits to halo density profiles using N-body simulations, showing that halo profiles are shallower than r^{-2} near the center and steeper than r^{-2} near the virial radius. The NFW profile is then a simple fitting formula to Equation (3), with a radially *varying* power-law $1 \leq \gamma(r) \leq 3$, to describe spherically averaged density profiles:

$$\rho(r) = \frac{\rho_0 r_s^3}{r(r+r_s)^2}. \tag{6}$$

Spherically integrating the NFW profile of Equation (6) one obtains a simple formula for the mass of dark matter,

$$M(r) = 4\pi\rho_0 r_s^3 \left\{ \ln(r+r_s) - \ln(r_s) - \frac{r}{r+r_s} \right\}, \tag{7}$$

which diverges logarithmically, for $r \gg r_s$. In the limit of small $r \ll r_s$, the NFW fitting formula of Equation (6) approaches the power-law with $\gamma \to 1$; and in the limit of large $r \gg r_s$ approaches the power-law with $\gamma \to 3$ – which does not approximate isothermal spheres. Navarro et al. [18] reported that rotation curves from galaxies ranging in size from giant to dwarf, satellites and gaseous atmospheres are compatible with the NFW

halo structure of Equation (6) provided the mass-to-light ratio increases with luminosity. Navarro et al. [18] determined that the central regions of the NFW distribution have densities comparable to the luminous parts of galaxies. Because there is a great amount of luminous stellar material in the cores of spiral galaxies and a deficit at large radii, the general consensus is that the dark matter halo component is not described by an isothermal sphere, but a model with a monotonically rising $\gamma(r)$ such that $\gamma(r) \ll 2$ inside the baryon dominated core, and $\gamma(r) \gg 2$ outside several Einstein radii.

Burkert [20] fitted a sample of several dark matter dominated dwarf galaxies employing a phenomenologically modified universal fitting formula,

$$\rho(r) = \frac{\rho_0 r_s^3}{(r+r_s)(r^2+r_s^2)}. \tag{8}$$

which, as in the case of the isothermal sphere of Equation (4), approximates a constant density core, $\gamma \to 0$ at $r \ll r_s$ – instead of a divergent $\gamma = 1$ core – but otherwise agrees with the NFW profile, with $\gamma \to 3$ at $r >> r_s$. Spherically integrating the Burkert model of Equation (8), one obtains an analytic fitting formula for the mass of dark matter,

$$M(r) = \pi\rho_0 r_s^3 \left\{\ln(r^2+r_s^2) + 2\ln(r+r_s) - 4\ln(r_s) - 2\tan^{-1}(r/r_s)\right\}, \tag{9}$$

which diverges logarithmically, for $r \gg r_s$.

Observationally, when including baryons, cores of galaxies and clusters of galaxies are better fitted by a generalized fitting formula to account for more or less cuspy cores. Zhao [21] hypothesized that the NFW fitting formula must be broadened to account for the basic observed features of galactic dynamics, including less cuspy cores:

$$\rho(r) = \frac{\rho_0 r_s^m}{r^n (r^\alpha + r_s^\alpha)^{(m-n)/\alpha}}, \tag{10}$$

where (α, m, n) are free parameters. The NFW fitting formula of Equation (6) corresponds to Equation (10) with an inner cusp with logarithmic slope $n = 1$, an outer corona with logarithmic slope $m = 3$, and a "turnover" exponent of $\alpha = 1$. Syer and White [22] argued that the existence of a $\gamma \ll 1$ core is inconsistent with the hierarchical formation scenario of dark halos, which are much more likely to result in cuspy central density distributions. The least cuspy fitting formula, the isothermal spheres of Equations (4) and (5) correspond to Equation (10) with a constant density inner core with logarithmic slope $n = 0$, an outer corona with logarithmic slope $m = 2$, and a "turnover" exponent of $\alpha = 2$. Although Burkert's fitting formula of Equation (8) cannot be expressed in the form of Equation (10), it does bridge the constant density, $\gamma \to 0$, core behavior of the isothermal sphere with the $\gamma \to 3$ large r behavior of the NFW profile.

Mechanisms involving angular momentum transfer and dynamical friction may work to flatten the cusp in the cores of galaxies and clusters by dissipating the energy of the baryonic component, possibly overwhelming the reciprocal effect of adiabatic compression [23, 24, 25, 26, 27, 28]. Sand et al. [29, 30] used a spectroscopic survey of gravitational arcs in a sample of clusters with stellar velocity dispersion data, and found shallower dark matter profiles in the central regions. Newman et al. [31] used weak and strong lensing data, coupled with resolved stellar velocity dispersion data, to provide a

detailed analysis of the baryonic and dark matter distribution in the lensing cluster Abell 611 between 3 kpc to 3.25 Mpc, and found shallower dark matter profiles in the central regions. Zitrin and Broadhurst [32], Zitrin et al. [33] used strong galaxy lensing around the most massive X-ray luminous cluster systems, producing surface density maps with shallow mass profiles due to relatively unconcentrated dark matter halos, with steeper profiles in the center due to the contribution of central galaxies.

A simple choice for the dark matter halo, corresponding to Equation (10), which allows for the observed baryon distributions, without adding to the cusp problem, and fits both galaxy rotations curves and X-ray cluster mass profiles across all scales [14], is the *Brownstein profile*:

$$\rho(r) = \frac{\rho_0 r_s^3}{r^3 + r_s^3}, \tag{11}$$

with a constant density inner core with logarithmic slope $n = 0$, an outer corona with logarithmic slope $m = 3$, and a turnover exponent of $\alpha = 3$. Without the core of baryons, Equation (11) approximates a constant density core, $\gamma \to 0$ at $r \ll r_s$ – instead of a divergent $\gamma = 1$ core – but otherwise agrees with the NFW profile, with $\gamma \to 3$ at $r >> r_s$. Spherically integrating the density profile of Equation (11), one obtains the *Brownstein fitting formula* for the mass of dark matter,

$$M(r) = \frac{4}{3}\pi\rho_0 r_s^3 \ln\left(1 + \frac{r^3}{r_s^3}\right), \tag{12}$$

which diverges logarithmically, for $r \gg r_s$. The concentration parameter,

$$c \equiv r_{200}/r_s, \tag{13}$$

where r_{200} is the radius at which the dark matter density is 200 times the Λ-CDM critical density, follows from

$$\frac{M(x)}{M_{200}} = \frac{\ln(1 + c^3 x^3)}{\ln(1 + c^3)}, \tag{14}$$

where $x = r/r_{200}$, M_{200} is the mass of dark matter enclosed by a sphere of radius r_{200}, and $M(x)$ is the mass of dark matter enclosed by a sphere of radius x in units of r_{200}.

The surface mass density for the profile of Equation (11) is calculated by projecting the radial coordinate, $r = \sqrt{R^2 + z^2}$, and integrating the density along the line-of-sight, z:

$$\Sigma(R) = 2\rho_0 r_s^3 \int_0^\infty \frac{dz}{(R^2 + z^2)^{\frac{3}{2}} + r_s^3}, \tag{15}$$

where R is in the plane perpendicular to the line-of-sight, z. The surface mass density is a maximum at $R = 0$:

$$\Sigma(0) = \frac{4\rho_0 r_s \sqrt{3}\pi}{9}, \tag{16}$$

and then drops off as

$$\Sigma(R) = \frac{2\rho_0 r_s^3}{R^2} \quad R > r_s. \tag{17}$$

Utilizing the form of the power-law of Equation (3), the power-law index of the profile of Equation (11) is minus the logarithm slope

$$\gamma(r) = -\frac{d\ln\rho(r)}{d\ln r} = \frac{3r^3}{r^3 + r_s^3}. \tag{18}$$

The central density, $\rho(0) = \rho_0$, is finite and may be written in terms of the Λ-CDM critical density, $\rho_c(z)$, and the dark matter central overdensity, δ_0,

$$\rho_0 = \rho_c(z)\delta_0, \tag{19}$$

where z is the redshift. Moreover, the dark matter density at $r = r_s$ is one-half the central density,

$$\rho(r_s) = \frac{1}{2}\rho_0, \tag{20}$$

and the power-law index of Equation (18) is

$$\gamma(r_s) = 3/2, \tag{21}$$

which is the intermediate value between the inner core with logarithmic slope $\gamma \to 0$, and outer corona with logarithmic slope $\gamma \to 3$. This means that the halo's constant density core is limited to the region $r < r_s$, where baryons dominate the galaxy, which is important for N-body simulations.

Although the baryons represent a small fraction of the total, including their distribution is dynamically important in the cores of galaxies and rich clusters, and the combination of visible baryons and the dark matter mass of Equation (12) produces a mass function which correctly fits galaxy rotation curves, and X-ray cluster mass profiles, as shown in §3.1 and §3.2, which is neither the case for the pseudo-isothermal sphere nor the NFW profile. Using cosmological N-body simulations, Davé et al. [34] showed that generalizations of the NFW profile, such as Equation (11), may alleviate the problems arising from the singular NFW profile due to a shallower core, and Wyithe et al. [35] showed that generalized gravitational lenses may resolve further observational discrepancies.

3. THE GLOBAL MASS TO LIGHT RATIO

The missing mass problem in galaxies and clusters of galaxies is best quantified by calculating the spherically averaged global mass to light ratio,

$$\Upsilon(r) = M(r)/L(r), \tag{22}$$

where $M(r)$ is the dynamically inferred mass inside a sphere of radius, r, and $L(r)$ is the observed luminosity integrated over the same region. We assume that the baryon mass to light ratio,

$$\Upsilon_{\text{baryon}} = M_{\text{baryon}}(r)/L(r), \tag{23}$$

is constant for a particular object, and so we may write the global mass to light ratio,

$$\Upsilon(r) = \Upsilon_{\text{baryon}}\Gamma(r), \tag{24}$$

in terms of the dynamic mass ratio:

$$\Gamma(r) \equiv M(r)/M_{\text{baryon}}(r). \tag{25}$$

The dynamic mass is computed according to the Newtonian force law of Equation (1), in which case a halo of dark matter is required to explain the missing mass:

$$M(r) = M_{\text{baryon}}(r) + M_{\text{halo}}(r), \tag{26}$$

as demonstrated in §3.1.1, or the dynamic mass is computed according to a particular modified theory of gravity according to Equation (25),

$$M(r) = \Gamma(r)M_{\text{baryon}}(r), \tag{27}$$

where the dynamic mass factor,

$$\Gamma(r) = G(r)/G_N, \tag{28}$$

depends on the particular modified gravity acceleration law:

$$a(r) = -G(r)M_{\text{baryon}}(r)/r^2. \tag{29}$$

The form of Equation (29) can be applied to any modified theory of gravity which deviates from the Newtonian inverse square law, including Milgrom's MOND [36, 37] as shown in §3.1.2, and Moffat's MOG [38, 39, 40, 41, 42, 43] as shown in §3.1.3.

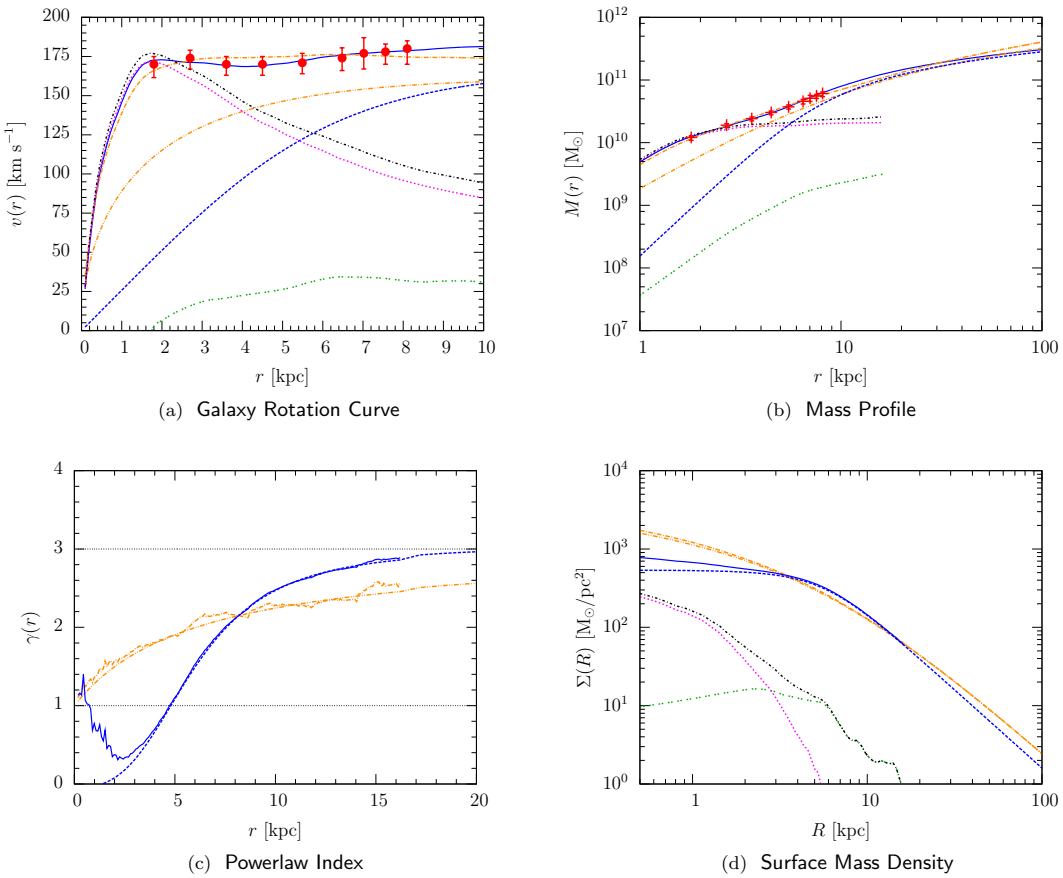
3.1. Galaxy Rotation Curves

3.1.1. Dark Matter

Every galaxy from the highest to lowest in surface brightness demonstrates a missing mass problem which increases with orbital distance from the center of the galaxy, and is greatest at the outermost extent of each galaxy, increasing monotonically with orbital distance. In Brownstein [14, Chapter 4], 29 HSB and LSB galaxy rotation curves were fitted with high-resolution Spitzer space telescope photometry using the dark matter fitting formula of Equation (11), producing excellent best fits to all of the galaxies including the *dwarf galaxies*, with consistent stellar mass to light ratios, showing a statistically significant reduction of the χ^2 per degree test of freedom in $\sim 90\%$ of the galaxies, over that of the NFW profile of Equation (6).

Figure 1a shows a comparison of galaxy rotation curves for the HSB galaxy UGC 6973, with the Brownstein model of Equation (11), shown in blue, and the NFW model of Equation (6) shown in yellow. All of the orphan features (kinks in the galaxy

v(r) data	red filled circles with error bars	HI gas	green dot-dotted line
M(r) data	red crosses with error bars	Stellar disk	magenta dotted line
Baryons (no Dark Matter)	black short-dash-dotted line		
Brownstein with Baryons	blue solid line	Brownstein halo	blue dashed line
NFW with Baryons	yellow dash-dotted line	NFW halo	yellow dash-dot-dotted line

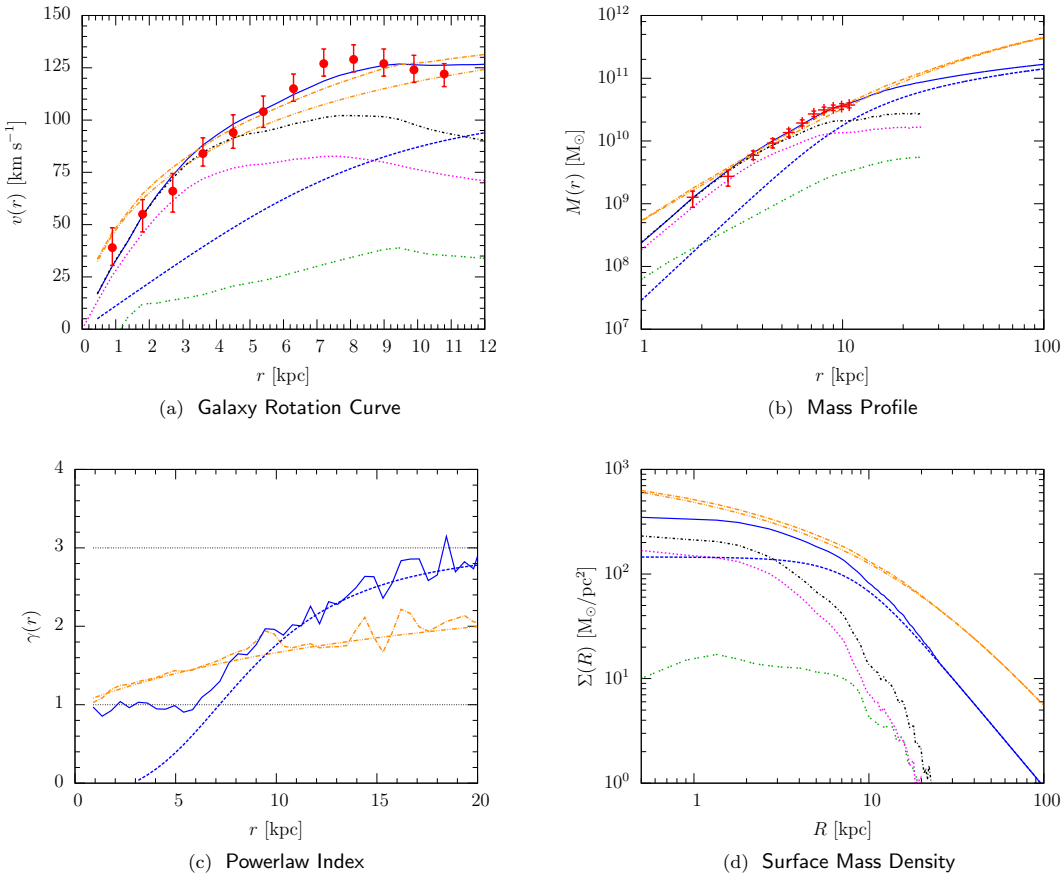


(a) Galaxy Rotation Curve (b) Mass Profile (c) Powerlaw Index (d) Surface Mass Density

FIGURE 1. High surface brightness (HSB) dwarf spiral galaxy UGC 6973 [14, Chapter 4].

rotation curve) were correctly repatriated to the luminous components. The integrated mass profiles, $M(r)$, demonstrate the transition between the baryon dominated core and the dark matter dominated halo, as shown in Figure 1b. Both models models produce logarithmically divergent mass functions. However, the Brownstein model requires a much less massive halo and allows for a physically reasonable stellar mass to light ratio, consistent across the entire sample. Every galaxy differed strongly from a single power-law density profile, $\rho \propto r^{-\gamma}$. However, the NFW model requires a significantly smaller stellar mass to light ratio, so that the variation in the logarithmic density slope, $\gamma(r)$ according to Equation (3), shown in Figure 1c, is approximately the same with or without baryons. Conversely, including the HI (and He) gas and stellar disk photometry led to a *baryon corrected* power-law index, $\gamma(r)$, varying between 1 (at small r, within

v(r) data	red filled circles with error bars	HI gas	green dot-dotted line
M(r) data	red crosses with error bars	Stellar disk	magenta dotted line
Baryons (no Dark Matter)	black short-dash-dotted line		
Brownstein with Baryons	blue solid line	Brownstein halo	blue dashed line
NFW with Baryons	yellow dash-dotted line	NFW halo	yellow dash-dot-dotted line



(a) Galaxy Rotation Curve (b) Mass Profile (c) Powerlaw Index (d) Surface Mass Density

FIGURE 2. Low surface brightness (LSB) dwarf spiral galaxy NGC 4010 [14, Chapter 4].

the baryon dominated core) and 3 (at large distances, within the dark matter halo). Surface mass density maps, $\Sigma(R)$, were produced for each galaxy as predictions for gravitational lensing studies, as shown for UGC 6973 in Figure 1d. Predictions for the Brownstein model differ from the NFW profile, which is everywhere dark matter dominated. However, like the NFW profile, the Brownstein model including baryons is nowhere flat.

Overall, the Brownstein model of Equation (11) provided very low χ^2 fits for each of the 19 HSB galaxies and each of the 10 LSB galaxies. This was not the case for the NFW model of Equation (6), which either fit the galaxy rotation curves very well, or failed to provide χ^2 best-fits for any positive definite stellar mass to light ratio. Figure 2a shows a comparison of galaxy rotation curves for the LSB dwarf galaxy NGC 4010, with the

Brownstein model of Equation (11), shown in blue, and the NFW model of Equation (6) shown in yellow. The Brownstein model repatriated all of the orphan features by requiring a stellar mass to light ratio consistent across the sample, whereas the NFW model required a nearly vanishing stellar mass to light ratio.

The integrated mass profiles, $M(r)$, demonstrate the transition between the baryon dominated core and the dark matter dominated halo, as shown in Figure 2b for the Brownstein model, whereas the the mass budget in the NFW model is everywhere dominated by dark matter. Remarkably, including the HI (and He) gas and stellar disk photometry led to a *baryon corrected* power-law index, $\gamma(r)$, for the Brownstein model, varying between 1 (at small r, within the baryon dominated core) and 3 (at large distances, within the dark matter halo), nearly identical to the NFW model which has completely suppressed the baryons by requiring a nearly vanishing stellar mass to light ratio, as shown for NGC 4010 in Figure 2c. As in the case of UGC 6973, the predicted surface density maps, $\Sigma(R)$, for gravitational lensing studies, as shown for NGC 4010 in Figure 2d show a baryon enhanced core for the Brownstein model, whereas the NFW model is everywhere dark matter dominated.

3.1.2. Milgrom's Modified Newtonian Dynamics (MOND)

The strong equivalence principle may be violated by preferred space-time fields. Milgrom [36] challenged the hidden mass hypothesis and introduced a nonrelativistic modification of Newtonian dynamics (MOND) at small accelerations, $a < a_0$, whereupon the gravitational acceleration of a test particle is given by

$$a\mu\left(\frac{a}{a_0}\right) = a_{\mathrm{N}}, \tag{30}$$

where $\mu(x)$ is a function that interpolates between the Newtonian regime, $\mu(x) = 1$, when $x \gg 1$ and the MOND regime, $\mu(x) = x$, when $x \ll 1$. Milgrom [37] introduced the interpolating function normally used for galaxy fitting,

$$\mu(x) = \frac{x}{\sqrt{1+x^2}}, \tag{31}$$

where

$$x \equiv x(r) = \left|\frac{\nabla\Phi(r)}{a_0}\right| = \left|\frac{a(r)}{a_0}\right|, \tag{32}$$

and determined that $a_0 \approx cH_0/6$, implying a cosmological connection to MOND:

$$a_0 = 1.0 \times 10^{-8}\ \mathrm{cm\ s^{-2}}. \tag{33}$$

Substituting Equations (31) and (32) into Equation (30) gives,

$$\frac{a(r)^2}{\sqrt{a(r)^2 + a_0^2}} = a_{\mathrm{N}}(r), \tag{34}$$

which has the solution,

$$a(r) = a_0 \sqrt{\frac{1}{2}\left(\frac{a_N(r)}{a_0}\right)^2 + \sqrt{\frac{1}{4}\left(\frac{a_N(r)}{a_0}\right)^4 + \left(\frac{a_N(r)}{a_0}\right)^2}}, \tag{35}$$

written in terms of the Newtonian acceleration of a test particle at a separation, r,

$$a_N(r) = \frac{GM(r)}{r^2}, \tag{36}$$

where $M(r)$ is the baryonic mass integrated within a sphere of radius, r. Substituting Equation (36) into Equation (30), the MOND acceleration law can be written,

$$a(r) = \frac{1}{\mu(r)} \frac{G_N M(r)}{r^2}, \tag{37}$$

and therefore MOND can be interpreted as gravity theory with a varying gravitational coupling according to Equation (29), with

$$G(r) = \frac{G_N}{\mu(r)}, \tag{38}$$

and a dynamic mass factor, as defined by Equation (28), with

$$\Gamma(r) = \frac{1}{\mu(r)}, \tag{39}$$

where $\Gamma(r) \sim 1$ in the Newtonian regime and $\Gamma(r) \gg 1$ in the MOND regime. It is important to note that MOND has a classical instability in the deep MOND regime corresponding to $\mu \to 0$ which leads to a divergent gravitational coupling, $\Gamma(r) \to \infty$, and that MOND violates the strong equivalence principle for all $\mu \neq 1$.

For gravity fields interior to galaxies and clusters of galaxies, the accelerations are sufficiently small that the MOND interpolating function, $\mu(x) \gg 1$, so that the Newtonian dynamic mass determined by MOND is much larger than the actual mass visible in the system, so that MOND may be an alternative to dark matter in these systems.

3.1.3. Moffat's Modified Gravity (MOG)

Clayton [44] showed that Moffat's massive nonsymmetric gravity theory (NGT) becomes identical to a Kalb-Ramond-Proca field [45] with an additional curvature coupling term when considered as a perturbation about a Ricci-flat background. Moffat [46, 47], Moffat and Sokolov [48] determined that in the weak-field approximation of the massive NGT relevant to galaxy dynamics, a range dependent Yukawa-type, fifth force [49] emerges in addition to the Newtonian $1/r^2$ central force; and asserted that this additional potential due to the interaction of the field structure with matter in the halos of

galaxies can explain galaxy rotation curves. van Nieuwenhuizen [50] found that the only massive antisymmetric tensor or vector fields free of ghosts, tachyons and higher-order poles in the propagator for linearized gravitation are the Maxwell-Proca fields. Isenberg and Nester [51] determined that only Maxwell fields, Proca fields, and purely longitudinal vector fields are free of instability when minimally coupled to gravity. Damour et al. [52] analyzed a class of physically consistent and ghost-free nonsymmetric gravity models with finite range massive gauge bosons, such as Moffat's massive NGT, and showed that the Kalb-Ramond-Proca field leads to minuscule – as yet unmeasured – deviations from Newtonian gravity at terrestrial scales consistent with stringent experimental bounds on possible violations, and the field may acquire *gravitational* strength at sufficiently large astrophysical scales.

Moffat [38] introduced Metric skew-tensor gravity (MSTG), where the modified acceleration law results from coupling the Kalb-Ramond-Proca field of the massive NGT to Einstein's metric. At astrophysical scales, the emergent low energy Yukawa meson is the only feature of the massive NGT left in MSTG to explain galaxy rotation curves.

Moffat [39] introduced the scalar-tensor-vector gravity (STVG) theory by adding a simpler, massive and self-coupled Maxwell-Proca field coupled to matter and gravity. The STVG effectively captures the fifth force due to a weak-field emergent Yukawa meson, simulating the predictions of massive NGT and MSTG, to leading order.

Moffat's modified gravity (MOG) theories have an acceleration law with a varying gravitational coupling according to Equation (29), with

$$G(r) = G_N\left\{1+\alpha\left[1-\exp(-\mu r)(1+\mu r)\right]\right\}, \tag{40}$$

and a dynamic mass factor, as defined by Equation (28), with

$$\Gamma(r) = \left\{1+\alpha\left[1-\exp(-\mu r)(1+\mu r)\right]\right\}, \tag{41}$$

In the absence of analytic solutions, MSTG requires phenomenological input from the Tully-Fisher relation, leading to the parameters, M_0 and r_0,

$$\alpha = \sqrt{\frac{M_0}{M}}, \tag{42}$$

$$\mu = 1/r_0, \tag{43}$$

used to fit galaxy rotation curves and X-ray cluster masses [40, 41, 42].

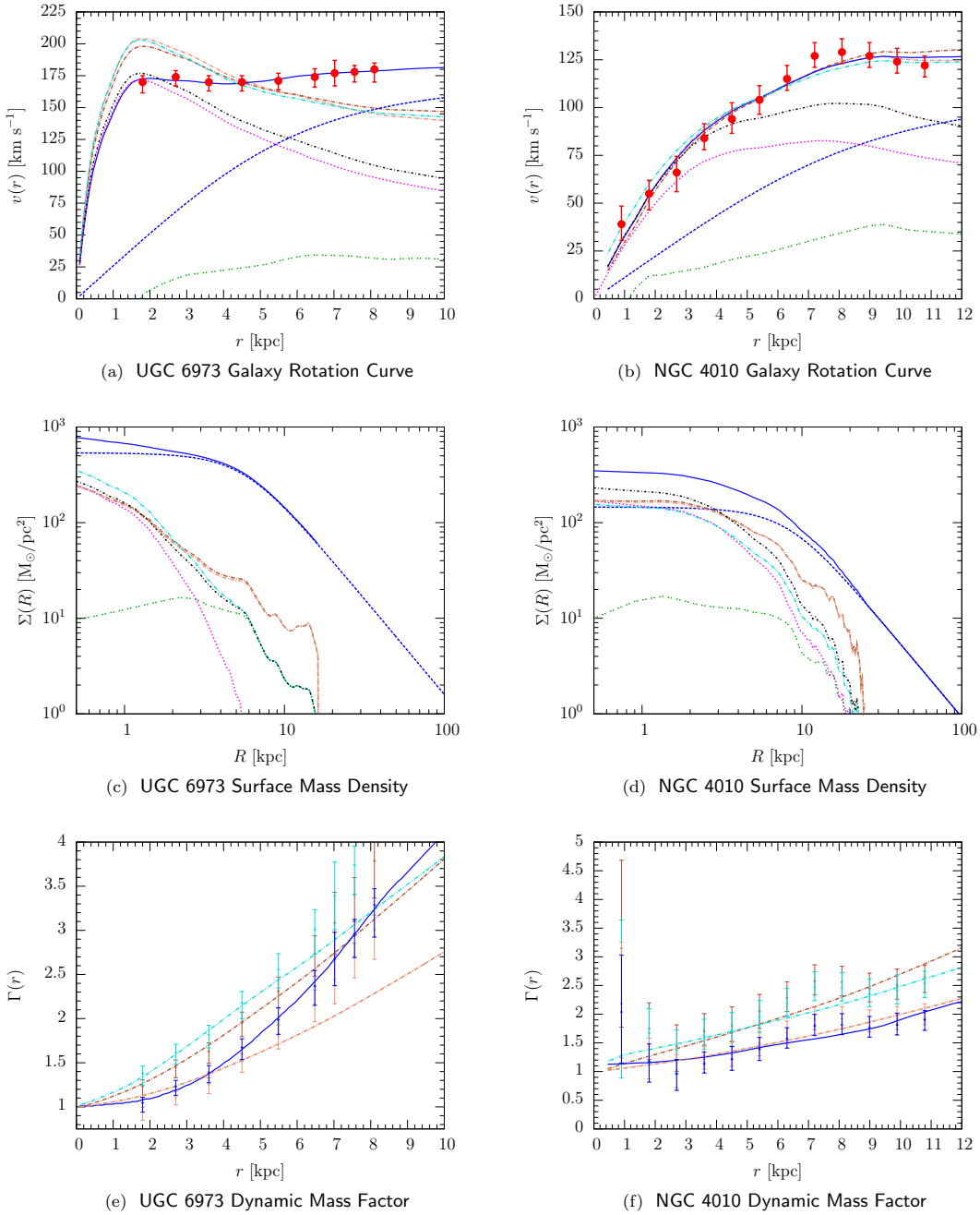
Moffat and Toth [43] showed that phenomenological input can be avoided in STVG by integrating the equations of motion in the weak-field, spherically symmetric limit, leading to solutions for α and μ as functions of the mass M,

$$\alpha = \frac{M}{\left(\sqrt{M}+E\right)^2}\left(\frac{G_\infty}{G_N}-1\right), \tag{44}$$

$$\mu = \frac{D}{\sqrt{M}}, \tag{45}$$

where the parameters, D, E and G_∞, are constants of integration to be determined.

v(r) data	red filled circles with error bars	HI gas	green dot-dotted line
Baryons (no Dark Matter)	black short-dash-dotted line	Stellar disk	magenta dotted line
Brownstein with Baryons	blue solid line	Brownstein halo	blue dashed line
Moffat's STVG	dark brown dash-dotted line	Moffat's MSTG	light brown dash-dotted line
Milgrom's MOND	cyan dash-dot-dotted line		



(a) UGC 6973 Galaxy Rotation Curve (b) NGC 4010 Galaxy Rotation Curve

(c) UGC 6973 Surface Mass Density (d) NGC 4010 Surface Mass Density

(e) UGC 6973 Dynamic Mass Factor (f) NGC 4010 Dynamic Mass Factor

FIGURE 3. Comparison of The Brownstein dark matter model to Milgrom's MOND, and to Moffat's MSTG and STVG theories for the high surface brightness (HSB) dwarf spiral galaxy UGC 6973 [left panels], and low surface brightness (LSB) dwarf spiral galaxy NGC 4010 [right panels] [14, Chapter 4].

Brownstein [14, Chapter 4] fitted a sample of 19 HSB and 10 LSB galaxies, comparing the Brownstein dark matter model to Milgrom's MOND, and Moffat's MOG.

To compute the MOND universal acceleration parameter, the MOND acceleration law of Equation (35) was applied with a variable a_0 yielding the subsample averages:

$$a_0 = \begin{cases} (1.34 \pm 0.66) \times 10^{-8}\ \mathrm{cm\ s^{-2}} & \texttt{HSB galaxy subsample,} \\ (1.02 \pm 0.78) \times 10^{-8}\ \mathrm{cm\ s^{-2}} & \texttt{LSB galaxy subsample.} \end{cases} \tag{46}$$

Because of the gross uncertainty in the averaged results, all galaxy rotation curves were refitted using Equation (33) universally.

The computation of the universal parameters in Moffat's MOG was performed by allowing M_0 and r_0 in MSTG, and D, E, and G_∞ in STVG to vary across the sample of 29 HSB and LSB galaxies, with the overall best-fitting results for each theory, respectively:

$$D = (6.4 \pm 0.2)\sqrt{M_\odot}\mathrm{pc}^{-1}, \quad E = (28.4 \pm 7.9) \times 10^3 M_\odot^{1/2}, \quad G_\infty = (24 \pm 18) G_N, \tag{47}$$

$$M_0 = (98.6 \pm 21.6) \times 10^{10} M_\odot, \quad r_0 = (16.4 \pm 6.1)\,\mathrm{kpc}. \tag{48}$$

The galaxy rotation curves were subsequently refitted as **one parameter** best-fits by varying only the stellar mass-to-light ratio, Υ, using Equations (47) and (48) universally.

Comparison of the best-fit galaxy rotation curves for UGC 6973, plotted in Figure 3a, shows that Moffat's MOG theories with universal parameters across the subsample perform as well as Milgrom's MOND theory, but not as well as the Brownstein dark matter model with variable dark matter halo parameters, ρ_0 and r_s. However, comparison of the best-fit galaxy rotation curves for NGC 4010, plotted in Figure 3b, shows that Milgrom's MOND and Moffat's MOG with universal parameters do as well as the Brownstein dark matter model, whereas the NFW model plotted in Figure 2b does not provide a comparable low χ^2 fit. Perhaps most significantly, the total mass with dark matter in the Brownstein model showed the least scatter of any of the Tully-Fisher relations, even when compared to Milgrom's MOND and Moffat's MSTG, which receive phenomenological input from the Tully-Fisher relation.

In order to better distinguish the models, it is necessary to perform precise strong gravitational lensing measurements of the total surface mass density, $\Sigma(r)$, plotted in Figure 3c and Figure 3d for UGC 6973 and NGC 4010, respectively.

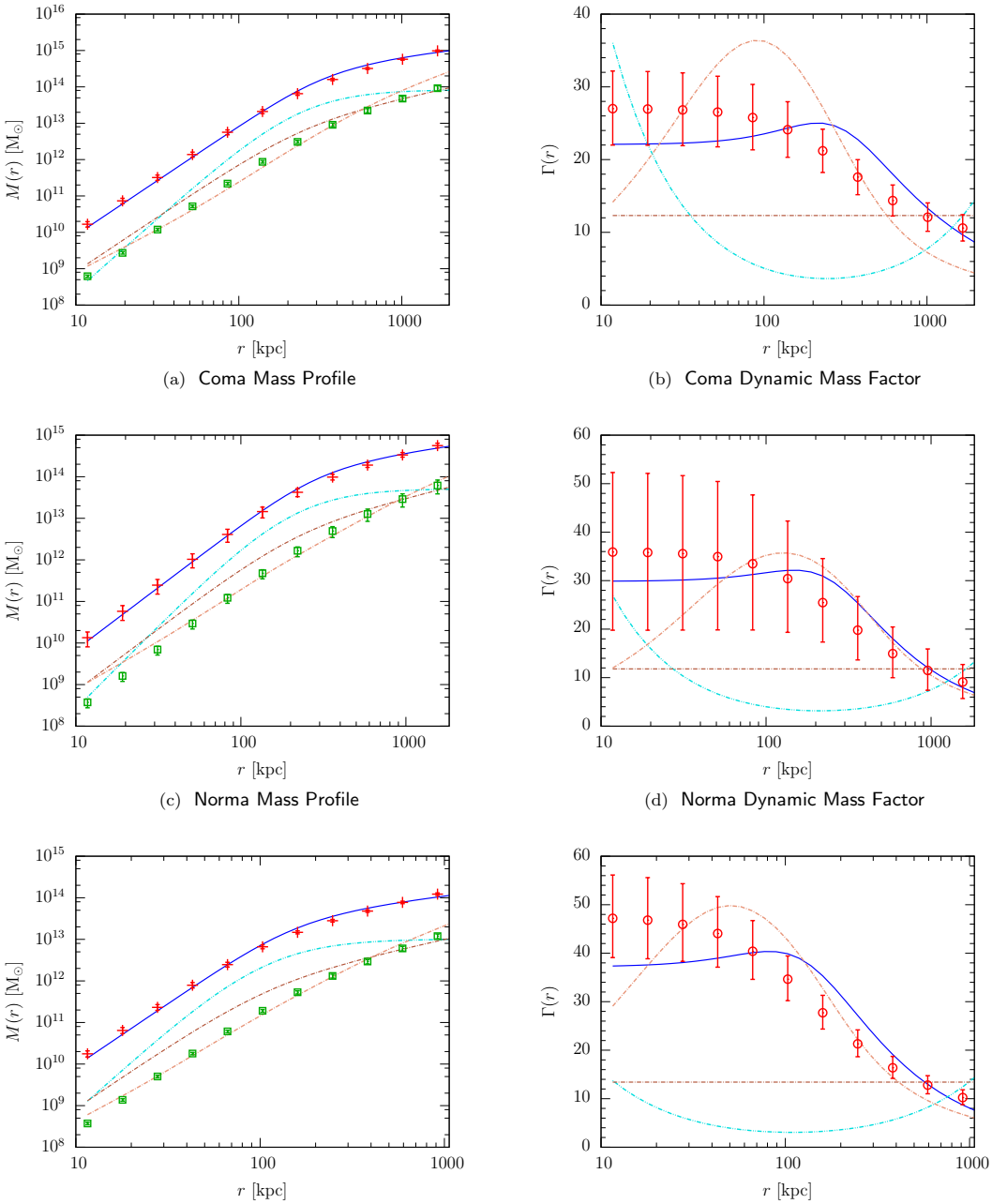
Nevertheless, it is remarkable that every model provides a similar picture in terms of the dynamic mass factor of Equation (25), plotted in Figure 3d and Figure 3e for UGC 6973 and NGC 4010, respectively. For the complete sample of 29 galaxies, the dynamic mass factor varies as,

$$\Gamma(r) \to 1 \text{ as } r \to 0, \tag{49}$$

$$3 \lesssim \Gamma_{\max}(r) \lesssim 7 \text{ as } r \to r_{\max}. \tag{50}$$

Therefore, we may conclude that every galaxy has a central core of baryons, outside of which there must be either a halo of dark matter, or a modified gravity region, which dominates the gravitational potential. This trend common among galaxy rotation curves is not seen in X-ray cluster masses, as detailed in §3.2, where the dynamic mass factor varies according to Equation (51), in contrast to Equations (49) and (50).

Newtonian dynamic mass	red crosses	Dynamic mass factor	red open circles
ICM gas mass	green open squares		
Brownstein with Baryons	blue solid line	Milgrom's MOND	cyan dash-dot-dotted line
Moffat's STVG	dark brown dash-dotted line	Moffat's MSTG	light brown dash-dotted line



(a) Coma Mass Profile (b) Coma Dynamic Mass Factor

(c) Norma Mass Profile (d) Norma Dynamic Mass Factor

(e) Abell 400 Mass Profile (f) Abell 400 Dynamic Mass Factor

FIGURE 4. Comparison of The Brownstein dark matter model to Milgrom's MOND, and to Moffat's MSTG and STVG theories for the Coma, Norma and Abell 400 X-ray clusters, showing mass profiles [left panels], and dynamic mass factors [right panels] [14, Chapter 5].

3.2. X-ray Cluster Imaging

Every X-ray emitting cluster of galaxies from 10 million degrees to 170 million degrees demonstrates a missing mass problem which is greatest at the center, and decreases with orbital distance from the center of the cluster, as seen by the variation in the dynamic mass factor of Equation (25), which varies as,

$$7 \lesssim \Gamma_{\max}(r) \lesssim 500 \text{ as } r \to 0, \tag{51}$$

and $\Gamma(r)$ decreases monotonically with orbital distance. Therefore, every cluster of galaxies is everywhere dominated by either dark matter, or modified gravity everywhere dominates the gravitational potential, particularly towards the center. Moreover, the missing mass problem is considerably stronger in clusters of galaxies, than galaxies.

Brownstein [14, Chapter 5] fitted a sample of 10 X-ray clusters of galaxies taken from a sample of 106 X-ray clusters [53, 54] considered in [41], comparing the Brownstein dark matter model of Equation (11) to Milgrom's MOND, and Moffat's MOG theories. The Brownstein dark matter model provided excellent fits with low χ^2 to every X-ray mass profile in the sample, as shown in Figure 4 for Coma, Norma and Abell 400, for example, whereas attempts to fit cluster mass distributions to NFW profiles led to large uncertainties due to a parameter degeneracy between the central density parameter, ρ_0, and the scale radius, r_s, which prevented the computation of the best-fit ρ_0 and r_s from converging, regardless of the χ^2. Without numerical convergence, the NFW results either over-predict the density at the core or under-predict the total mass.

Although MOND predicts convergent mass profiles, none matched any of the observed distributions to low χ^2. Treating the MOND universal acceleration, a_0, as a variable parameter did little to improve the χ^2 because the dynamic mass factor predicted by MOND at the scale of clusters of galaxies does not correctly track the data, as shown in the rightmost panels of Figure 4, whereas the Brownstein dark matter model follows the data across the entire cluster within one standard deviation, for every cluster of galaxies, tracking the data with a large, and nearly constant, $\Gamma(r)$, which decreases monotonically at large radii.

Moffat's MSTG was not able to provide low χ^2 fits to any of the clusters of galaxies with the universal parameters of Equation (48), determined from fitting the galaxy rotation curves. In order to better determine the scale dependence of the parameters, it is reasonable to treat the MSTG mass and range parameters as variable and to perform two-parameter best-fits to the X-ray gas masses of the sample of clusters of galaxies. MSTG was able to provide moderate to low χ^2 fits to all of the X-ray mass profiles by allowing variable M_0 and r_0, as shown in the leftmost panels of Figure 4. The dynamic mass factors in MSTG, plotted in the rightmost panels of Figure 4, are similar to the Brownstein model at large distances from the cluster center, but do not track the data with the observed constant Γ needed to fit the core. Whereas tabulation of the best-fit MSTG parameters provided no statistical support for universal constants, there was strong statistical support that the MSTG parameters are scale-dependent.

Moffat's STVG was not able to provide low χ^2 fits to any of the clusters of galaxies with the universal parameters of Equation (47), determined from fitting the galaxy rotation curves. Whereas every weak fitting STVG one-parameter best-fit by a variable

stellar mass-to-light ratio, Υ, show dramatic improvement and reduction in the reduced χ^2/ν statistic using a four-parameter best-fit including variable parameters, the tabulation of D, E, and G_∞ provided no statistical support that the STVG integration constants are universal. However, for values of D sufficiently large and values of E sufficiently small, the STVG gravitational coupling and dynamic mass factor of Equations (40) and (41), respectively, simplify to their asymptotic form,

$$G(r) = G_\infty, \quad \Gamma(r) = G_\infty/G_N, \tag{52}$$

and are independent of r. Therefore, in order to determine the scale dependence of the STVG asymptotic coupling, it is reasonable to treat G_∞ as variable and to perform one-parameter best-fits to the X-ray gas masses of the sample of clusters of galaxies, shown in the leftmost panels of Figure 4. Because the STVG dynamic mass factor is constant, whereas the data is only constant in the core and then monotonically decreasing thereafter, the spherically symmetric STVG model does not correctly track the missing mass problem in clusters of galaxies, as shown in the rightmost panels of Figure 4.

4. CONCLUSIONS

The Brownstein model of Equation (11) does provide a universal fitting formula from which excellent fits, with low χ^2, are obtained across the scales from dwarf galaxies to the largest cluster of galaxies, provided the observed baryons are not neglected in the mass budget. This is an important step in imaging dark matter halos, particularly because the NFW model of Equation (6) fails at the low mass scale of dwarf galaxies, and at the high mass scale of X-ray clusters of galaxies. Conversely, none of the alternatives to dark matter considered in [14] fit the missing mass problem with common universal parameters across the scales from dwarf galaxies to clusters of galaxies. This suggests that if the equivalence principle is violated at astrophysical scales with sufficient strength to explain the missing matter problem, then the process must be dynamical, allowing for running coupling constants, explaining the success of Moffat's MOG to fit the data at all scales with variable parameters.

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